

ePBL Internship Report

CampusHub: A Full Stack Web-Based College Management System

Domain: Full Stack Development

Submitted By

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1 Abstract

CampusHub is a web-based College Management System developed using PHP, MySQL, HTML, CSS, Bootstrap, and JavaScript. The application provides separate portals for administrators, faculty members, and students. It simplifies academic management by providing secure authentication, student and faculty management, attendance tracking, CGPA management, and a centralized notice board. The system enhances administrative efficiency by centralizing academic information, reducing manual work, and providing a secure, responsive, and user-friendly platform for effective college management.

2 Introduction and Objectives

Educational institutions handle a vast amount of information related to students, faculty members, academic performance, attendance, and administrative activities. Traditional methods of maintaining these records manually are time-consuming, prone to errors, and difficult to manage as the number of students increases. A centralized digital platform helps institutions improve efficiency, reduce paperwork, and provide quick access to academic information.

CampusHub is a Full Stack Web-Based College Management System developed to simplify the management of academic records through a secure and user-friendly web application. The system provides separate portals for administrators, faculty members, and students, each with role-based access to perform their respective tasks. It enables administrators to manage student and faculty records, faculty members to update academic information such as CGPA and attendance, and students to view their academic details and important notices.

The project was developed using modern web technologies including PHP, MySQL, HTML, CSS, Bootstrap, and JavaScript. It demonstrates the practical implementation of Full Stack Web Development concepts such as frontend design, backend programming, database integration, session management, authentication, and CRUD (Create, Read, Update, Delete) operations. The application provides a responsive interface and ensures efficient communication between all stakeholders within the institution.

The objectives of the project are:

- Develop a secure and user-friendly College Management System.
- Implement role-based authentication for Admin, Faculty, and Students.
- Manage student and faculty records efficiently using CRUD operations.
- Maintain and update student attendance percentage and CGPA.
- Provide a centralized notice board for important announcements.
- Demonstrate practical implementation of Full Stack Web Development technologies.
- Improve data organization, accessibility, and overall administrative efficiency.

3 System Architecture

CampusHub follows a three-tier architecture consisting of Presentation, Application and Database layers.

- Presentation Layer: HTML, CSS, Bootstrap and JavaScript.
- Application Layer: PHP business logic and session management.
- Database Layer: MySQL stores application data.

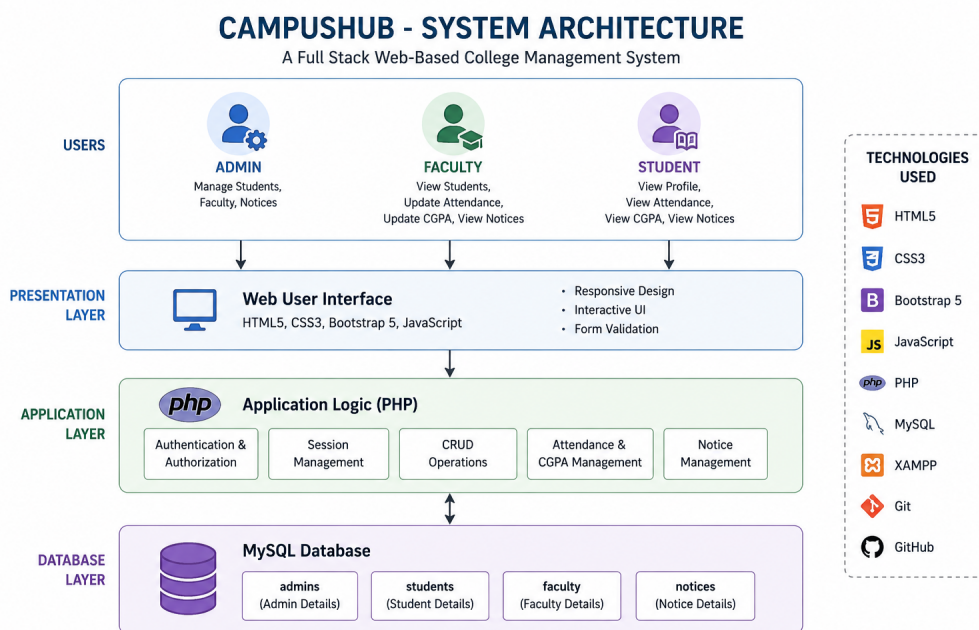


Figure 1: System Architecture

CampusHub was developed as a Full Stack Web-Based College Management System using PHP, MySQL, HTML, CSS, Bootstrap, and JavaScript. The application follows a role-based architecture, where three types of users—Administrator, Faculty, and Student—can securely access the system according to their assigned permissions. The implementation focuses on providing a centralized platform for managing academic records while ensuring security, usability, and efficient database operations.

The frontend was designed using HTML5, CSS3, and Bootstrap to create a clean, responsive, and user-friendly interface. Bootstrap components such as cards, tables, navigation bars, and buttons were used to maintain consistency across all pages. JavaScript was utilized for client-side interactions and improving the user experience.

The backend was implemented using PHP, which processes user requests, performs authentication, manages sessions, validates user input, and executes CRUD (Create, Read, Update, Delete) operations. PHP communicates with the MySQL database through MySQLi to retrieve and update information securely. Session management ensures that only authenticated users can access their respective dashboards.

The MySQL database serves as the central repository for storing administrator, faculty, student, attendance, CGPA, and notice information. Database operations were

designed to maintain consistency while providing fast retrieval and efficient management of records.

Admin Module

The Administrator module serves as the central control panel of the application. After successful authentication, the administrator is redirected to a dashboard displaying statistical information such as the total number of students, faculty members, and administrators.

The administrator can perform the following operations:

- Secure login using administrator credentials.
- Add, edit, delete, and search student records.
- Add, edit, delete, and search faculty records.
- Publish and delete notices displayed to all users.
- View overall system statistics through the dashboard.
- Manage the complete academic database.

This module provides complete administrative control over the CampusHub application.

Faculty Module

The Faculty module provides a secure environment for faculty members to manage academic information. Faculty members have limited access based on their role, ensuring that administrative functions remain protected.

The Faculty module provides the following features:

- Secure faculty login.
- View student information.
- Update student CGPA.
- Update student attendance percentage.
- View notices published by the administrator.
- Logout securely after completing operations.

Faculty members are intentionally restricted from adding or deleting faculty accounts or student records, maintaining proper role-based access control.

Student Module

The Student module provides students with access to their academic information through a personalized dashboard. Students can securely log in using their credentials and monitor their academic progress.

The Student module includes:

- Secure student login.
- View personal profile information.
- View CGPA.
- View attendance percentage.
- View notices published by the administrator.
- Secure logout functionality.

Students have read-only access to their information, ensuring the integrity and security of academic records.

Notice Board Module

CampusHub includes a centralized Notice Board that allows the administrator to publish important announcements. The latest notices are automatically displayed on the dashboards of administrators, faculty members, and students.

This module simplifies communication within the institution by eliminating the need for physical notice boards and ensuring that all users receive the same information through the application.

Authentication and Security

The application implements secure login functionality using PHP session management. Each user must authenticate using valid credentials before accessing the system. After successful login, users are redirected to their respective dashboards based on their role.

Unauthorized access is prevented by verifying active sessions on every protected page. Users who attempt to access restricted pages without authentication are redirected to the login page.

Version Control

The complete CampusHub project was managed using Git and GitHub. Version control was used to track project progress, maintain source code, document the application through a detailed README, and provide secure project hosting. The repository also includes the database export, screenshots, project report, and setup instructions, making the application easy to review and deploy.

4 Results

The application was successfully implemented and tested.

- Secure role-based login
- Student and Faculty management
- Attendance management
- CGPA management
- Notice board
- Responsive dashboards
- MySQL integration
- GitHub version control

Project Screenshots

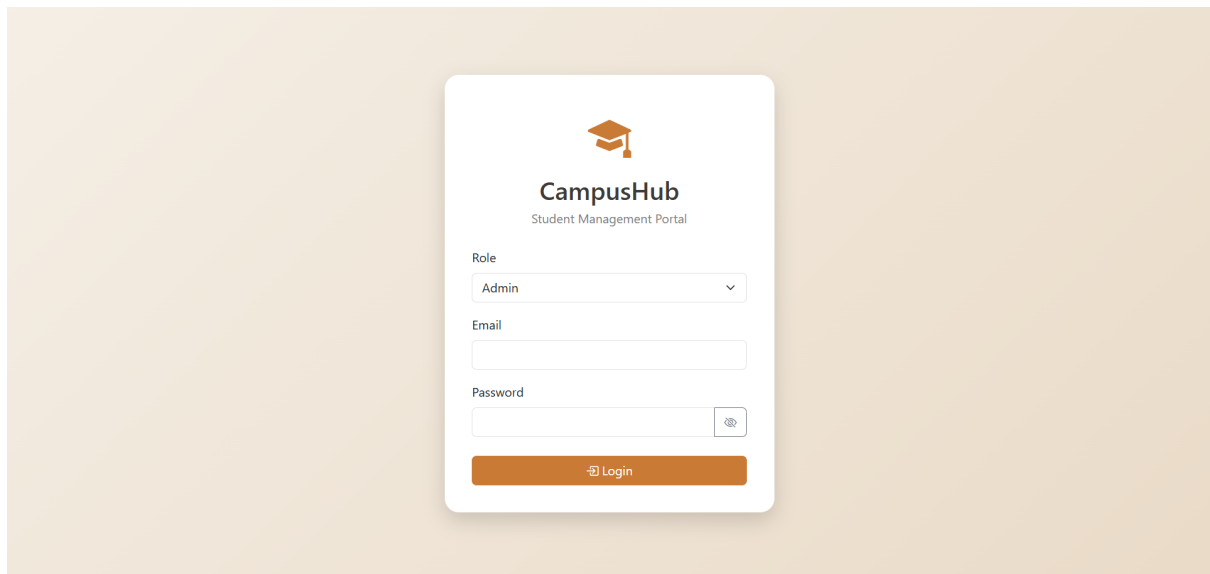


Figure 2: Login Page

The login page provides secure authentication for all users of the CampusHub system. Administrators, faculty members, and students can log in using their respective credentials and are redirected to their role-specific dashboards after successful authentication.

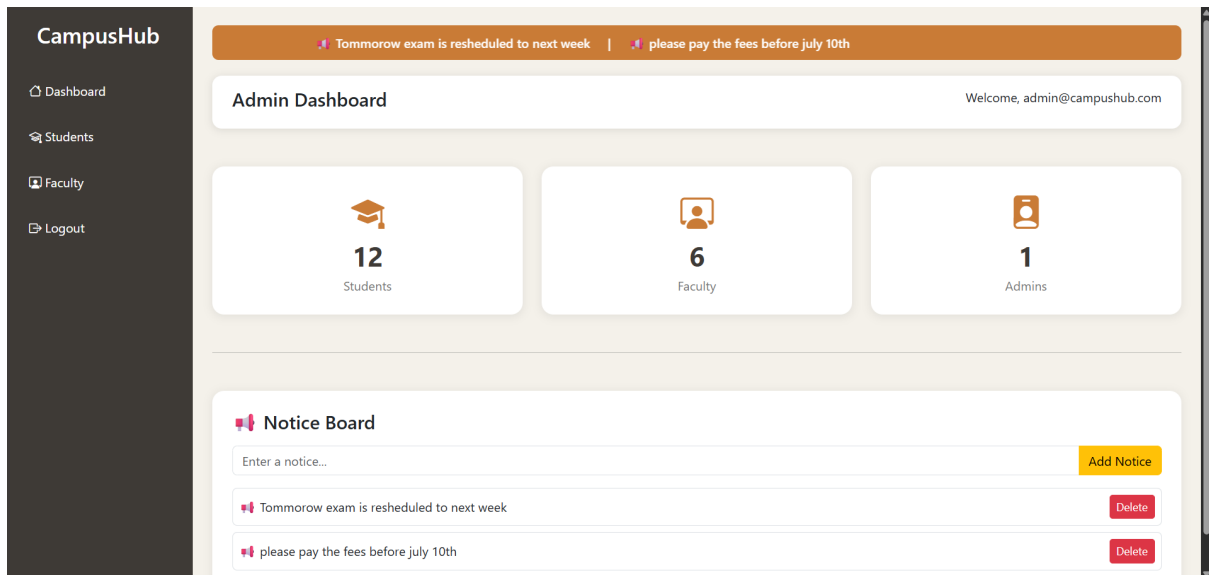


Figure 3: Admin Dashboard

The Admin Dashboard acts as the central control panel of the application. It displays overall statistics such as the total number of students, faculty members, and administrators while providing quick access to all management modules.

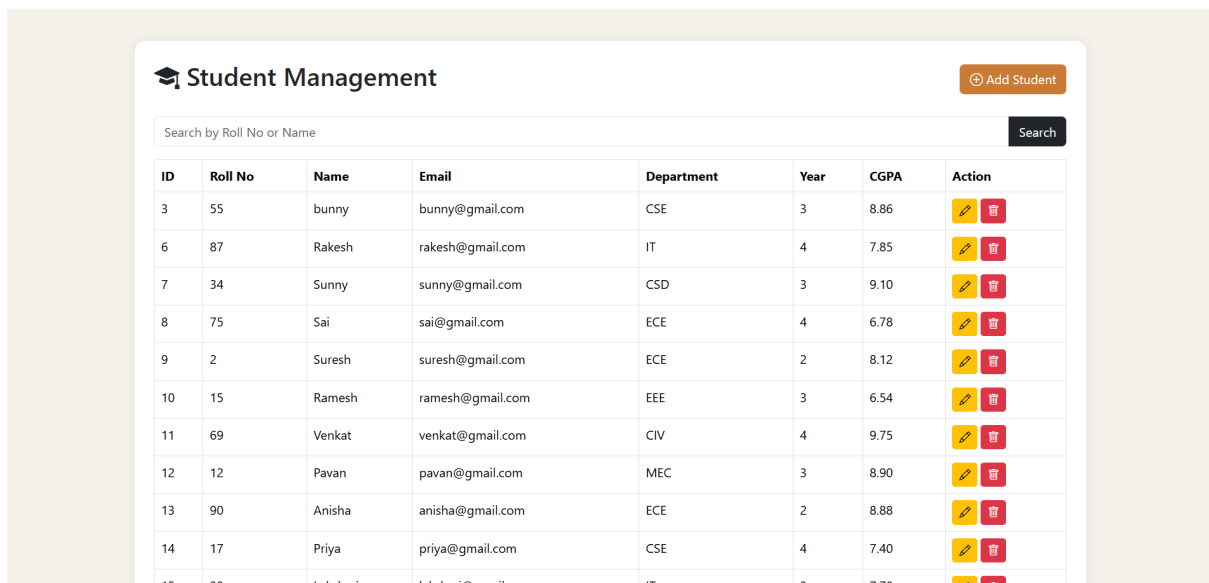


Figure 4: Student Management

The Student Management module enables the administrator to add, edit, delete, and search student records. It simplifies the maintenance of student information through an organized and user-friendly interface.

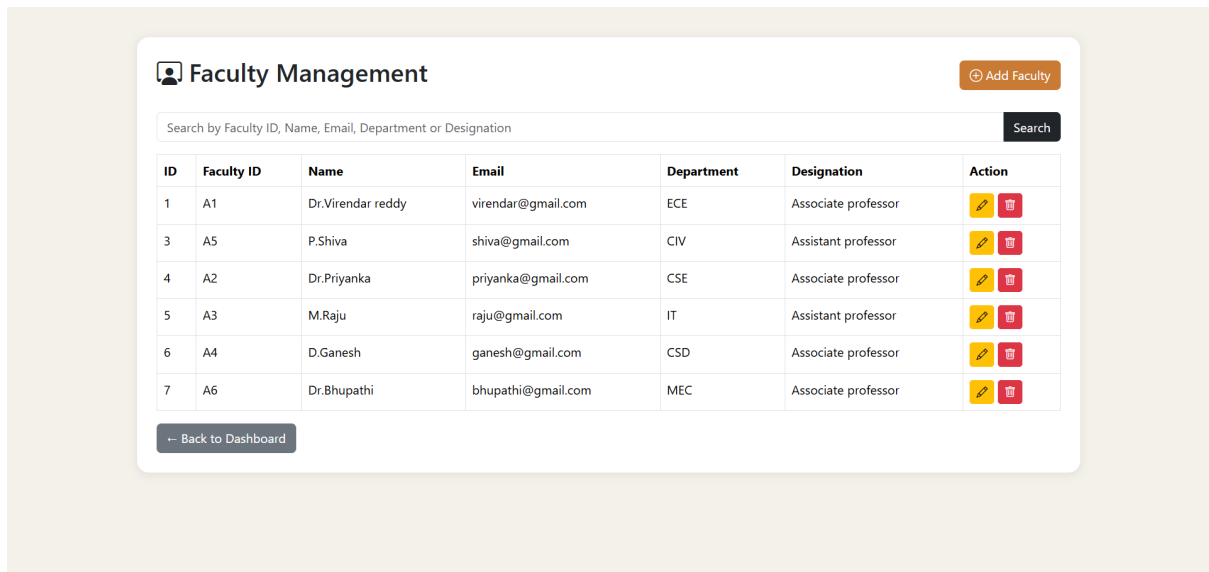


Figure 5: Faculty Management

The Faculty Management module allows administrators to efficiently manage faculty information. It supports adding new faculty members, updating existing records, deleting records, and searching faculty details.

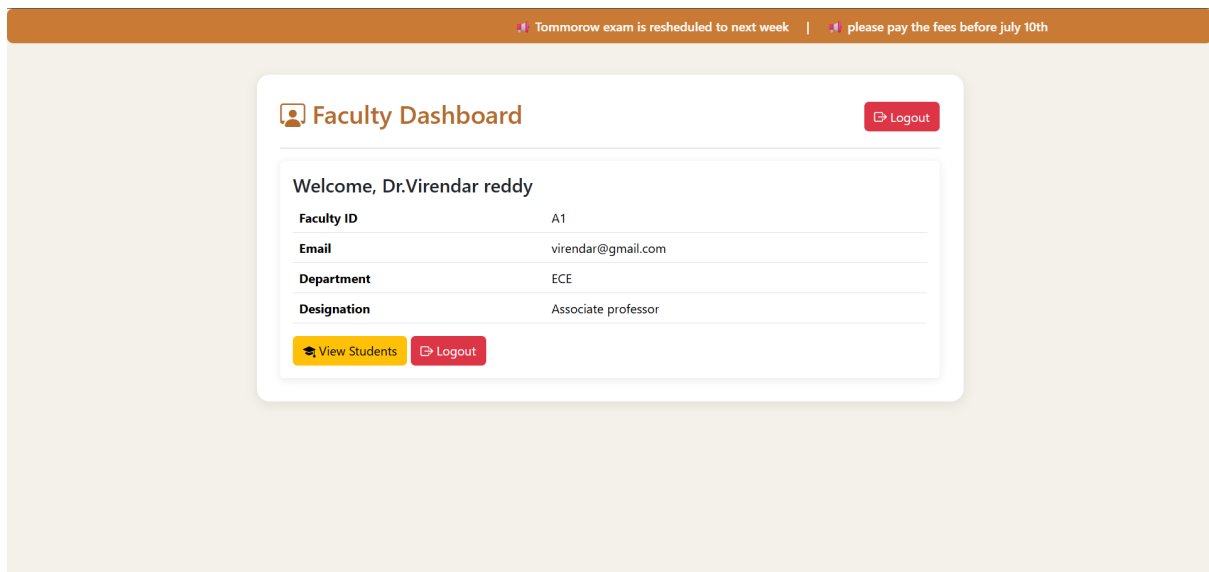


Figure 6: Faculty Dashboard

The Faculty Dashboard provides faculty members with access to student information, attendance percentage, CGPA updates, and notices. It offers a simple interface for managing academic records assigned to faculty.

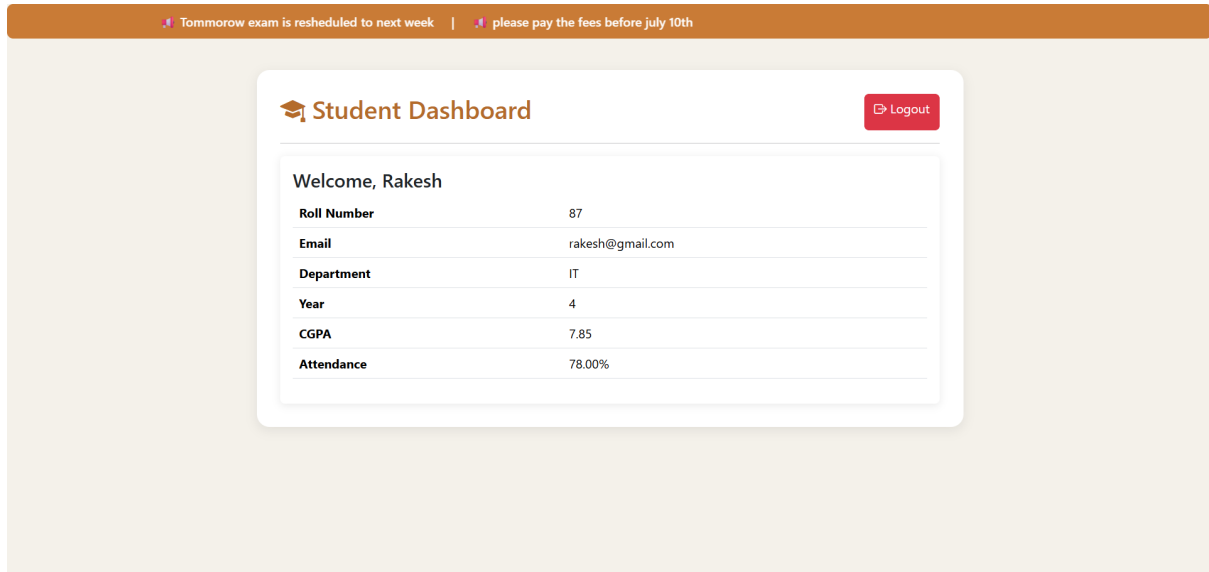


Figure 7: Student Dashboard

The Student Dashboard displays the student's personal academic information, including attendance percentage, CGPA, and important notices published by the administration. It provides read-only access to ensure data security and integrity.

5 Conclusion and Future Enhancements

CampusHub successfully demonstrates the implementation of a Full Stack Web-Based College Management System using PHP, MySQL, HTML, CSS, Bootstrap, and JavaScript. The application provides secure role-based access for administrators, faculty members, and students, enabling efficient management of academic records, attendance, CGPA, and notices. The project enhanced practical knowledge of frontend development, backend programming, database management, and version control using Git and GitHub.

Future enhancements include:

- Password encryption for improved security.
- Email notification system.
- Profile image upload.
- PDF report generation.
- Analytics dashboard with charts.
- Cloud deployment.
- Mobile application support.

6 References

References

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